



Financial technology and financial capability: Study of the European Union

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ARTICLE INFO

Keywords:

Financial capability
Financial technology
Financial well-being
Personal finance
European Union
JEL: D14, G51, G53, O52

ABSTRACT

Enhancing household financial capability is important for mitigating severe economic challenges. In the European Union (EU), financial technology (FinTech) solutions are considered critical for managing household finance, but their role in enhancing financial capability is ambiguous. Herein, we measure financial capability in the EU and investigate the effect of FinTech using three waves of panel data from Global Findex (2014, 2017, and 2021) and Eurostat Databases. According to the analyses, EU countries vary greatly in terms of FinTech and financial capability, with countries in the Union's north scoring remarkably high. Results emphasize a considerable effect of FinTech on financial capability and highlight an increase in the latter when the Human Development Index increases. Practical guidelines for measuring financial capability are also presented to assist countries that require additional efforts to effectively address financial capability challenges.

1. Introduction

Research findings regarding household financial capability are a valuable resource for policymakers, providing insights that can aid in effectively planning a nation's household financial landscape. Although studies in the United States and Canada show high levels of achievement, such research appears to be rare in the European Union (EU). Major household financial surveys, such as the European Household Finance and Consumption Survey, have not considered financial capability, resulting in a lack of knowledge about household financial capability in Europe. Furthermore, exploring these issues poses several challenges because the research diverges in how it measures financial capability (Birkenmaier, Rothwell, and Agar, 2022).

Sherraden et al. (2022) conclude that low financial ability prevents individuals from owning property, arguing that financially capable people will experience less financial stress during economic downturns. Xiao and Bialowolski (2023) recently challenged dominant research by emphasizing the need to explore the factors critical to fostering sustained financial capability while focusing on the consequences of achieving such capability. Among the few attempts to examine critical factors, French et al. (2020) investigate the influence of smartphone app use on financial behavior and suggested a positive impact of financial technology (FinTech) on financial capability in the UK. In the United States, Lusardi et al. (2021) emphasize the benefits of using online technology to make financial decisions. Meanwhile, FinTech has gained popularity in the EU, with a significant percentage of households using these applications on

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a daily basis (Statista, 2024). FinTech enables individuals to track their financial transactions, schedule payments, and manage savings plans (Nourallah & Öhman, 2021), which can improve their financial capability.

This study is based on the capability approach, developed by Amartya Sen, the winner of the 1998 Nobel Memorial Prize in Economics, and used by other important contributors such as Lusardi (2011) to identify variables to consider when measuring financial capability. Another aim is to investigate the relationship between FinTech and financial capability, taking into account control variables. The study makes use of three waves of panel data from the Global Findex (2014, 2017, and 2021) (Demirgüç-Kunt et al., 2022), and Eurostat (Eurostat, 2024) databases.

This study, which measures five variables: skills, debt, saving, financial resilience, and financial well-being, identifies the top three EU countries in terms of financial capability, namely Sweden, the Netherlands, and Austria, as well as the countries with the lowest financial capability scores. Furthermore, the study finds that Hungary and Latvia have experienced the greatest percentage growth in financial capability. Sweden, Finland, and Estonia top the FinTech rankings. Despite Romania and Bulgaria's modest FinTech scores, they appear to be on the right track, as they have the highest percentage growth in FinTech in the EU.

The findings show that FinTech has a significant positive effect on financial capability, which remains stable even after several endogeneity and robustness checks. The study also found that the Human Development Index has a significant positive effect on the treatment of financial capability. However, the study found no significant relationship between income and financial capability. Financial freedom is also considered limited in an individual's financial capability.

To ensure the robustness of the results, the study employs subsample techniques, dividing the countries under study into two subsamples based on their level of implementation of the Europe 2020 strategy. The complementary results confirm FinTech's significant positive effect on financial capability, with countries adhering closely to the Europe 2020 strategy outperforming countries in the second subsample. The latter may be related to the findings of Sánchez and Ruiz-Martos (2018), who found that households in countries with high scores for the Europe 2020 strategy tend to have higher life satisfaction.

The study makes three significant contributions. First, to the best of the authors' knowledge, this is the first study of household financial capability in the EU following the development of measures of five financial capability variables, namely skills, debt, savings, financial resilience, and financial well-being. Using these variables to capture various aspects of financial capability can help overcome the relatively naive prior approach to measuring this capability and pave the way for future studies. Second, this study contributes to increased knowledge of the overall financial capability situation in the EU by investigating the current financial capability of European countries, concluding that countries in northern Europe have a significantly higher level. Third, by highlighting areas of deficient FinTech in the EU, this study contributes to the Digital Finance Strategy, which aims to improve the use of FinTech solutions in the EU while reducing digital service fragmentation between countries (European Banking Authority, 2024). The present study can be linked to two projects managed by the European Central Bank (2024). The Digital Europe Project aims to provide EU households with secure digital payment methods and will benefit from the FinTech assessment results. The study also suggests that the Household Finance and Consumption Survey include a financial capability section to align with the US national financial capability study (FINRA Investor Education Foundation, 2023) and the Canadian financial capability survey (Government of Canada, 2023).

The remainder of the paper is structured as follows. Section 2 provides theoretical background, and Section 3 presents data and the empirical model. Section 4 reports the results, and Section 5 concludes with a discussion and policy recommendations.

2. Theoretical background

2.1. Financial capability

Financial capability roots in Amartya Sen's seminal work (Johnson & Sherraden, 2007) and the capability approach, which focuses on individuals' ability to address various "functionings" important to their lives (Sen, 2005). Functioning is a crucial notion in Sen's approach because it allows people to manage a variety of activities. In a primitive context, functionings may concern having enough food, whereas in more sophisticated situations, it can mean complex actions (Sen, 1993). This approach is distinct in that it focuses on actions rather than passivity. Furthermore, the ability to choose is critical in the Sen approach. In this regard, Nussbaum (2011) compares two individuals: the first is "fasting" and the other is "starving." She argues that, although both individuals lack the food they require, they do not have the same opportunities because the fasting person chose not to eat, whereas the starving person did not have food to eat. Sen (1993) defines a capability as "a set of such functioning n-tuples, representing the various alternative combinations of beings and doings any one (combination) of which the person can choose."

Regarding financial capability, the World Bank (2013) states that it "encompasses the knowledge, attitudes, skills, and behaviors of consumers concerning managing their resources and understanding, selecting, and making use of financial services that fit their needs." The importance of an active rather than passive approach to financial capability has been highlighted in previous studies in this field (French et al., 2020). The former is evident in Lusardi's (2011, p.2) definition of financial capability: "how well people make ends meet, plan ahead, choose and manage financial products, and possess the skills and knowledge to make financial decisions." Xia and Tao (2020, p. 15) emphasize the active approach, concluding that financial capability is the "consumer ability to apply appropriate knowledge and perform desirable financial behavior to achieve financial well-being."

2.2. Measuring financial capability

Birkenmaier, Rothwell, and Agar (2022) scrutinize how earlier studies measured financial capability and find that many studies used various methods to assess this type of capability. Their results show that most previous studies took a naive approach to measuring financial capability, relying primarily on financial access and knowledge. As a result, a more innovative approach is required to consider the factors that are critical when investigating financial capability in developed countries (Birkenmaier, Maynard, and Kim, 2022; Xiao et al., 2022).

Reading and numerical skills are required for making sound financial decisions (Hastings et al., 2013). Following Sen's approach, households with financial skills are likelier to take advantage of opportunities such as purchasing property and managing portfolios. It is important to note that skills in this context differ from knowledge, which is defined as "the ability to understand personal and broader financial matters, [while] 'skill' is the ability to apply that knowledge in everyday life" (French et al., 2020, p. 305).

Debt is typically considered when studying household financial planning and management issues (Almenberg et al., 2021). According to Lusardi (2011), it is critical to consider both assets and debt when assessing financial capability. Debt can affect household financial capability because high debt can lead to financial vulnerability and difficulty meeting financial obligations (Klapper & Lusardi, 2020). Property debt, vehicle loans, and student loans are popular among households in developed countries (Zinman, 2015). In 2021, the household debt level² as a percentage of net disposable income was 255% in Denmark, 222% in the Netherlands, and 203% in Sweden. Thus, considering all types of debt when determining financial capability is likely to produce misleading results. To address this issue, we rely on Xia and Tao (2020, p. 15), who argue that access to informal debt or exclusion from traditional financial services is a reasonable indicator of low financial capability.

Saving is essential for households (Tufano, 2009), and households that save money can manage financial shocks without incurring additional debt (Brounen et al., 2016). Such households can invest their savings and earn dividends, which can help them achieve financial independence (Browning & Lusardi, 1996). Financial resilience, or dealing with unexpected financial expenses, is another aspect of financial capability. Households that save money and manage their debt will be better able to handle unexpected financial expenses (Tahir et al., 2022).

Not only money-related issues can help households become financially capable; psychological factors are also crucial (Financial Consumer Agency of Canada, 2019). According to prior studies (Taylor et al., 2011), households with strong financial capabilities are less likely to experience money-related physiological issues. Due to its role in mitigating financial anxiety and increasing financial security (Barrafrem et al., 2020), financial well-being seems an irreplaceable aspect when financial capability is considered.

Taken together, skills, debt, savings, financial resilience, and financial well-being all appear to be important factors in determining financial capability.

2.3. Financial technology and financial capability

Although its origins can be traced back to early attempts to use analog technology in the financial industry (Nguyen et al., 2023), embryonic FinTech can be found in the 15th century, when people conducted financial transactions with physical media such as papers and coins (Alt et al., 2018). However, advanced technology, such as blockchain, has enabled the development of customized, fast, and secure FinTech solutions (Gomber et al., 2018; Lee et al., 2023). In 2019, global investments in FinTech solutions exceeded USD 135 billion, and as a result of the widespread use of these solutions by households, several central banks around the world are considering issuing novel types of money, such as central bank digital currencies (Keister & Sanches, 2023).

The reasons for widespread adoption of FinTech solutions extend beyond their ease of use and reasonable prices. FinTech solutions, such as mobile banking apps, can assist people in making sound financial decisions (Nourallah & Öhman, 2021). Furthermore, financial robo-advisors enable investors to easily build portfolios and track their investments (Kuvvet, 2022). These robo-advisors can reduce behavioral biases and help investors gain financial knowledge and learn about financial planning (Nourallah, 2003). Various online financial information sources have become popular among households, with personal finance blogs regarded as critical to household financial decisions (Hoffmann & Otteby, 2018).

Taken together, households that use FinTech solutions can compare products, read reviews written by others, and may be able to use artificial intelligence solutions such as ChatGPT. They can also use various online financial resources to learn more about financial issues, which will likely improve their financial situation (Panos & Wilson, 2020).

3. Data and empirical model

3.1. Data

In June 2022, the World Bank released the fourth wave of the Global Findex Database (Demirgüç-Kunt et al., 2022). Panel data from 2011, 2014, 2017, and 2021 cover various aspects of household access to and use of financial services, and have previously been used in household finance studies (Demir et al., 2022; Sha'ban et al., 2020). Eurostat is the statistical office that provides statistics for the EU (Eurostat, 2024). The Eurostat database section contains statistics on the economy and finance, population and social

² Household debt is defined as "all liabilities of households (including nonprofit institutions serving households) that require payments of interest or principal by households to the creditors at a fixed dates in the future" (Organization for Economic Co-operation and Development, 2023).

conditions, among other variables. Previous studies have used these datasets to address household financial issues in the EU (Nuzzo & Piermattei, 2020). Unfortunately, some figures from the Global Findex Database were unavailable for 2011. Therefore, we must eliminate the first wave while keeping the remaining three. We also exclude Luxembourg, Slovenia, and the Slovak Republic due to a lack of data. For the 2014, 2017, and 2021 waves, we used balanced panel data from 24 European countries.³

3.2. Variables

3.2.1. Financial capability

Financial capability is measured by using the arithmetic means of the five constructs, as shown in Eq. (1).

$$\text{Financial capability} = \text{skills} + \text{debt} + \text{saving} + \text{financial resilience} + \text{financial wellbeing}/5 \quad (1)$$

The measurements are as follows (variables, definitions and their sources are provided in Appendix 1):

- “Skills” is measured by the “participation rate in education and training” (Eurostat, 2024).
- The arithmetic means of the percentage of respondents who stated that they borrowed (i) for health or medical purposes (% age 15+) and (ii) from family or friends (% age 15+) are calculated (Demirgüç-Kunt et al., 2022). Higher percentages indicate lower financial capability. Hence, we use 1 minus the arithmetic means of the previous percentages to find the debt variable.
- In line with Demirgüç-Kunt et al. (2022), saving is measured as the arithmetic mean of (i) saving any money (% age 15+), (ii) saving for old age (% age 15+), and (iii) saving at a financial institution (% age 15+).
- Financial resilience is measured by the ability to face unexpected financial expenses, which is calculated as 1 minus the inability to face unexpected financial expenses (Eurostat, 2024).
- Financial well-being is measured using (i) the average satisfaction rating and (ii) the distribution of the population aged 18+ with a health status considered very good (Eurostat, 2024).

3.2.2. Financial technology

Previous studies apply “using mobile payment for paying bills” as a proxy for FinTech use (Demir et al., 2022). To properly measure FinTech, this study takes into account variables that target various aspects of FinTech, specifically the arithmetic means of the following four variables: (i) received digital payments (% age 15+), (ii) made a digital payment (% age 15+), (iii) used a mobile phone to pay utility bills (% age 15+), and (iv) individual frequency of Internet use (Appendix 2).

3.2.3. Control variables

According to the financial competence framework for adults in the EU, the European Union-Organization for Economic Co-operation and Development (2022) state that “income-related competencies are essential for financial resilience and relevant for many other competencies such as budgeting, saving and retirement.” Households with a high income have a higher level of financial literacy (Lusardi et al., 2017), whereas households with a low income have a lower level of savings (Almenberg et al., 2021). Furthermore, “financial freedom is an indicator of banking efficiency as well as a measure of independence from government control and interference in the financial sector” (Heritage Foundation, 2022a). Hence, financial freedom can help determine financial capability and thus improve the efficiency of the EU’s financial sector (Chortareas et al., 2013). Recent studies (Ofosu-Mensah Ababio et al., 2021; Zeqiraj et al., 2022) have also taken the Human Development Index into account when studying issues related to household financial capability, and Collins and Urban (2020) report an interrelation between financial freedom and financial well-being. Thus, three control variables are included: received wages, financial freedom, and the Human Development Index. In addition, we replace received wages with the annual percentage of gross domestic product (GDP) per capita growth, i.e., GDP per capita growth (annual %) (Appendix 2).

We include additional control variables to ensure that the effects of the key explanatory variable do not overlap with those of other variables. By improving the business environment, these additional variables (Appendix 2) may indirectly affect financial capability. Given the rule of law in promoting growth in wealthy countries (Aixalá & Fabro, 2008), we use data from the World Justice Project (2021) to account for any such effects. Trade openness (i.e., international openness) may also affect household financial position because more foreign products in a country may encourage households to spend more money on consumption (The World Bank, 2022). When focusing on the impact of FinTech on income inequality, Demir et al. (2022) find that trade openness has a negative effect. Furthermore, we take into account the use of technology in the business sector, specifically companies that use software solutions, because extensive digitalization may affect household financial capability (Iddrisu & Chen, 2022; Van Bon, 2021). Increased stock market returns are also likely to increase financial capability (Van Rooij et al., 2011).

³ Panel data have time-series elements, so the unit root test is required for testing stationarity. This study employs the test of Levin, Lin, & Chu, 2002, which fits the balanced panel data and is equivalent to the Augmented Dickey–Fuller (ADF) test. The Levin-Lin-Chu (LLC) test assumes that error terms are independently distributed across panels and follows a stationary invertible autoregressive moving-average process for each panel; i.e., the null hypothesis is that all the panels contain a unit root, if the significant result indicates that the panels are stationary. However, for LLC, the standard panel procedures may not be computationally feasible or sufficiently powerful for panels of less than moderate size, i.e., N between 10 and 250 and T between 25 and 250 (Baltagi & Baltagi, 2008). Stationarity testing is not an essential prerequisite for this study. Relying on unit roots in the case of a small value of T may be misleading, particularly when no long-term relationship exists.

3.3. Descriptive statistics

Fig. 1 shows that Sweden, the Netherlands, and Austria had the highest financial capability scores in 2021: 0.684, 0.656, and 0.652 out of 1, respectively. Bulgaria, Romania, and Greece had the lowest scores: 0.447, 0.462, and 0.447, respectively (Appendix 3).

Fig. 2 shows that Hungary and Latvia had the highest percentage growth in financial capability in the EU from 2014 to 2021, at 34.3% and 25.5%, respectively, while Denmark experienced slightly negative growth (1.1%). Overall, the EU had a growth rate of 11.5% in financial capability between 2014 and 2021.

Fig. 3 shows that Sweden, Finland, and Estonia had the highest FinTech scores in 2021, with 0.868, 0.823, and 0.809 out of 1, respectively. The lowest scores were found in Romania and Bulgaria at 0.518 and 0.533, respectively (Appendix 4).

According to Fig. 4, Greece, Romania, and Bulgaria experienced the greatest percentage growth in FinTech from 2014 to 2021: 101.4%, 82.1%, and 61.4%, respectively. All countries experienced positive growth in the FinTech index throughout the period, with growth rates increasing in the second phase (2017–2021).

Table 1 presents descriptive statistics for the minimum, first quartile, median, mean, standard deviation, third quartile, and maximum values. The mean and median values are very similar and sometimes identical. Therefore, there is no issue with data distribution. The variance inflation factors were calculated, and none of the corresponding values exceeded the critical value of 10, indicating no multicollinearity issues.

3.4. Model specification

The following baseline regression model is estimated, with X representing the control variables in Eq. (2), to determine whether FinTech can address financial capability in EU countries and to examine the role of control variables.⁴

$$\text{Financial capability}_{i,t} = \alpha_1 + \alpha_2 \text{financial technology}_{i,t} + \sum_{n=1}^n \rho_n X_{n,i,t} + u_{i,t} \quad (2)$$

3.5. Estimation strategy

The fixed-effects approach considers country-specific effects while eliminating unobserved heterogeneity between countries (Baltagi & Baltagi, 2008). This estimation only considers within-individual differences, effectively discarding any information about differences between individuals.

Although the fixed-effects estimator accounts for the effects of time-invariant variables, it cannot estimate them. The within-country transformation removes the time-invariant variables, so we cannot estimate their coefficients. Thus, employing a random-effects strategy may produce better results. The estimator uses both within- and between-country variances to estimate the effect of time-invariant variables, resulting in a more efficient estimator. However, using this strategy necessitates strict exogeneity and orthogonality between regressors and specific fixed effects. Hence, we used the Hausman (1978) test. The null hypothesis implies the advantage of the random-effects model, whereas the alternative is the fixed-effects model (Greene, 2008). The Chi-square of the Hausman test is $\chi^2 = 20.03$ with a significant p -value (i.e., $p < 0.01$). We therefore reject the null hypothesis, indicating a correlation between the unique errors and the regressors. This also confirms the fixed-effects strategy's advantage.

3.6. Serial correlation and heteroskedasticity

A panel data approach necessitates some specification tests for serial correlation (common in panel data) and heteroscedasticity. Serial correlation biases standard errors, reducing the efficiency of estimates. The classical linear panel data model assumes that disturbances are serially uncorrelated. This assumption may be violated because the dynamic effect of shocks on the dependent variable is frequently spread across multiple periods (Baltagi & Baltagi, 2008). Most serial correlation tests rely on specific assumptions about the nature of individual effects. Wooldridge (2022) proposes a test with few assumptions. Using this test, we obtained an F statistic of 8.131 and rejected the null hypothesis at a 1% significance level. This shows that we do not rule out the existence of serial correlation. Therefore, we cluster the standard errors by country.

Because the model may not capture all differences between the countries under study, we account for the heteroskedasticity that is likely to appear in the data. A modified Wald statistic, which is viable when the assumption of normality is violated at least in asymptotic terms (Greene, 2008), shows that $\chi^2 = 31,227.37$ with a significant p -value. This implies that error terms have heteroskedastic variances across groups. Accordingly, we use robust standard errors in our regression analyses.

4. Results

4.1. Econometric results

Ordinary least squares (OLS) and fixed-effects results show that FinTech is positively and significantly related to financial capability

⁴ $u_{i,t} = u_i + v_{i,t}$, where u_i denotes unobservable individual-specific effects and $v_{i,t}$ denotes the reminder disturbance (Baltagi & Baltagi, 2008).

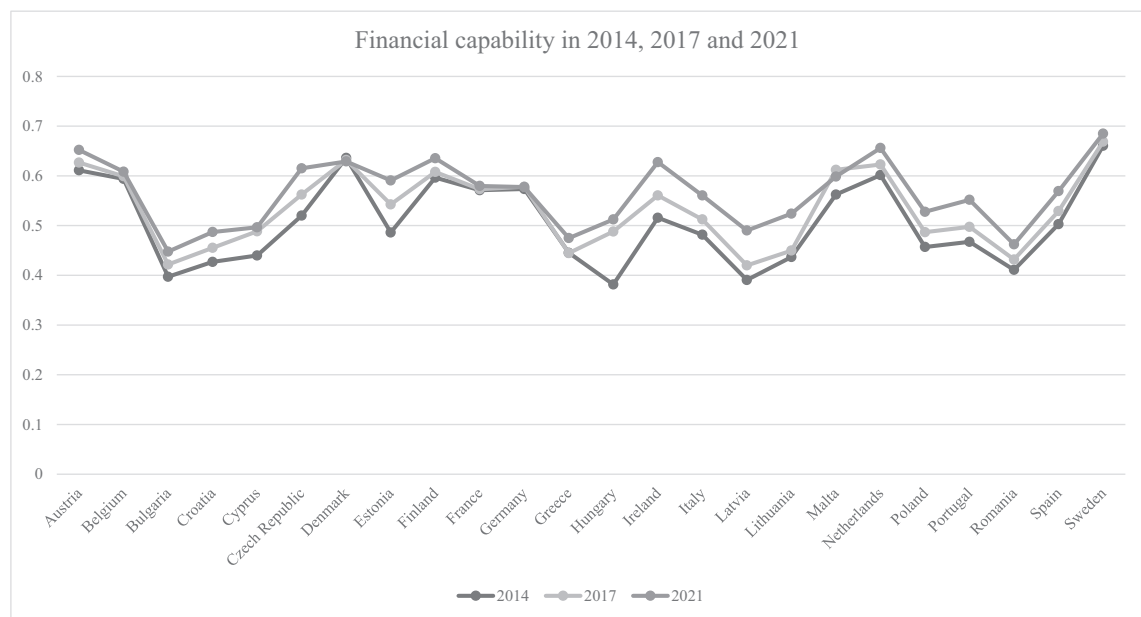


Fig. 1. Financial capability.

Note: Fig. 1 presents the level of financial capability on three years, 2014, 2017, and 2021. Data is based on three waves of panel data (2014, 2017, and 2021) of the Global Findex and the Eurostat Databases. The 24 countries represent members of the EU, i.e., all countries except Luxembourg, Slovenia, and Slovak Republic.

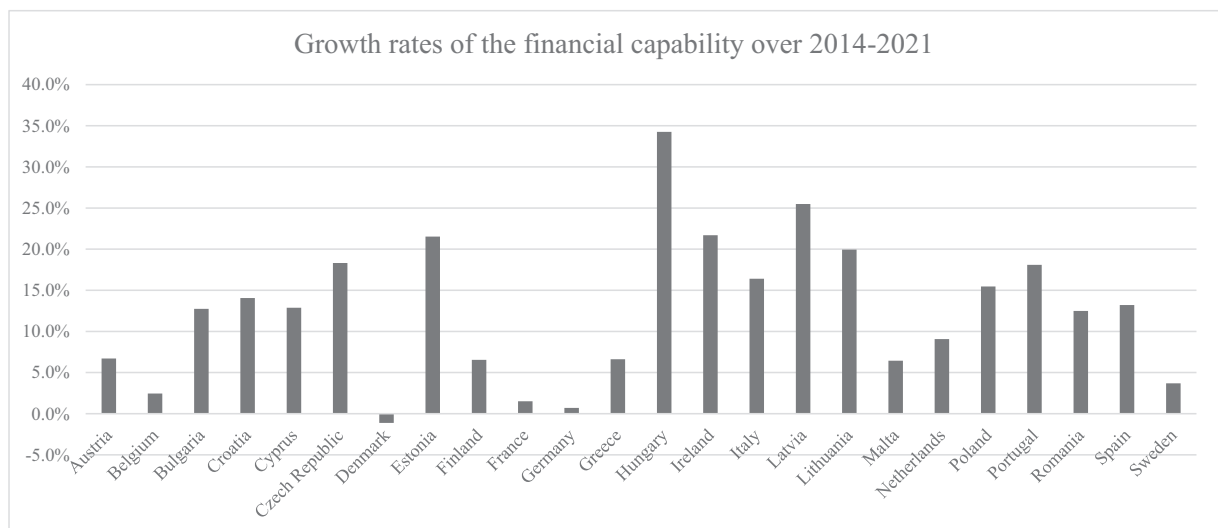


Fig. 2. The growth of financial technology.

Note: Fig. 2 presents the percentage of growth of financial capability on between 2014 and 2021. Data is based on three waves of panel data (2014, 2017, and 2021) of the Global Findex and the Eurostat Databases. The 24 countries represent members of the EU, i.e., all countries except Luxembourg, Slovenia, and Slovak Republic.

at the 1% level (see Table 2). Model (1) displays the results of the OLS test, whereas Models (2) through (4) display the results of fixed-effects models, one variable at a time. Model (5), which includes all variables, shows an R-squared value of 0.64. This finding is consistent with the conclusion of D'Acunto and Rossi (2023) that some FinTech solutions, such as robo-advisors, can assist households in avoiding the consequences of behavioral biases by providing the necessary information, allowing them to make informed financial decisions.

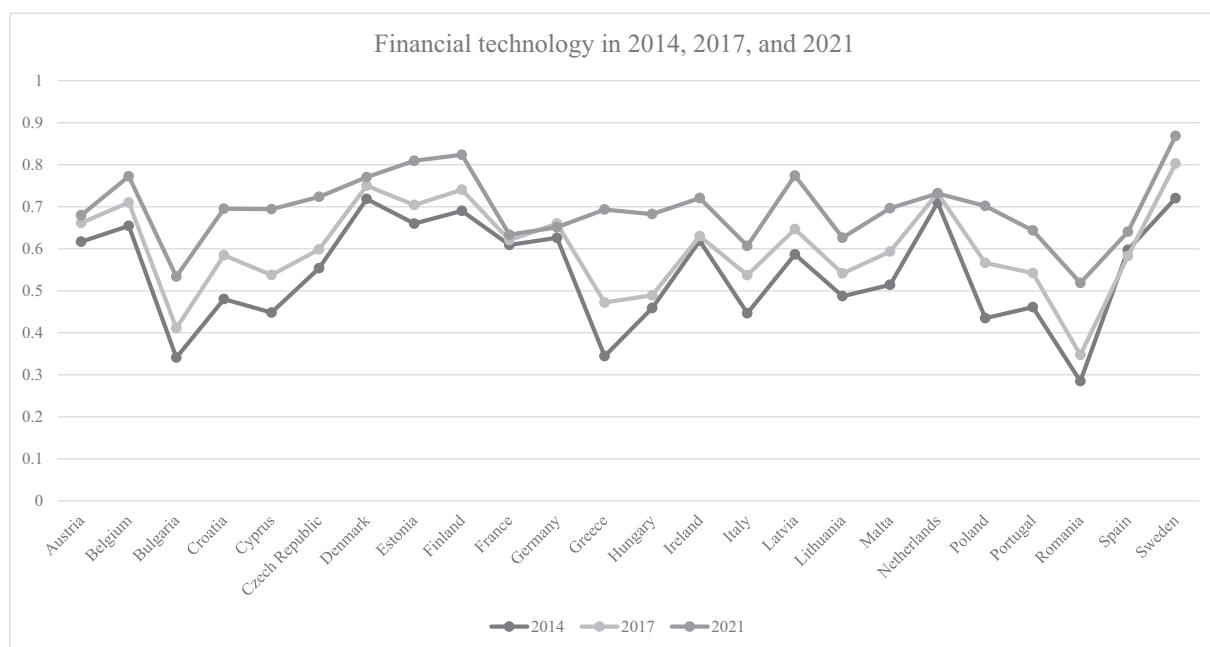


Fig. 3. Financial technology.

Note: Fig. 3 presents the level of FinTech on three years, 2014, 2017, and 2021. Data is based on three waves of panel data (2014, 2017, and 2021) of the Global Findex and the Eurostat Databases. The 24 countries represent members of the EU, i.e., all countries except Luxembourg, Slovenia, and Slovak Republic.

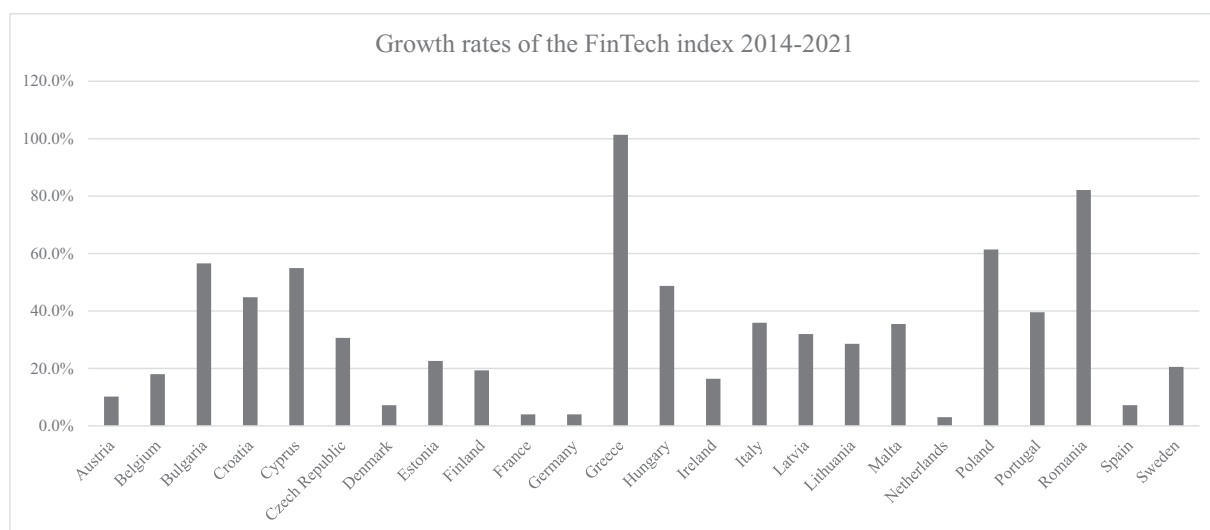


Fig. 4. The growth of financial technology.

Note: Fig. 4 presents the percentage of growth of FinTech between 2014 and 2021. Data is based on three waves of panel data (2014, 2017, and 2021) of the Global Findex and the Eurostat Databases. The 24 countries represent members of the EU, i.e., all countries except Luxembourg, Slovenia, and Slovak Republic.

The results also show a significant positive effect of the Human Development Index on financial capability. This is consistent with [Sha'ban et al. \(2020\)](#), who found a significant effect of the Human Development Index on financial inclusion. However, the present study found no significant relationship between income (i.e., received wages) and financial capability, nor between financial freedom and financial capability. These results contradict previous studies indicating that financial literacy increases with income ([Klapper &](#)

Table 1
Descriptive statistic.

	Minimum	First quartile	Median	Mean	Standard deviation	Third quartile	Maximum
Financial capability	0.381	0.471	0.535	0.535	0.078	0.600	0.685
Financial technology	0.285	0.537	0.628	0.614	0.123	0.702	0.869
Received wages	0.290	0.457	0.525	0.518	0.091	0.579	0.709
Financial freedom	0.550	0.663	0.700	0.699	0.056	0.741	0.814
Human Development Index	0.795	0.866	0.892	0.892	0.039	0.931	0.948

The table presents the descriptive statistics of the variables used in the econometric models where financial capability is the dependent variable and the FinTech is the independent variable while the three other variables (Received wages, Financial freedom, and Human Development Index) are the control variable. Data are based on three waves of panel data (2014, 2017, and 2021) from the Global Index (Demirgüç-Kunt et al., 2022) and the Eurostat Databases (Eurostat, 2024) and includes 24 European countries, i.e., all countries except Luxembourg, Slovenia, and Slovak Republic.

Table 2
Estimations of financial capability equations.

	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
VARIABLES	OLS	FE1	FE2	FE3	FE4
FinTech	0.206*** (0.0694)	0.311*** (0.0493)	0.334*** (0.0565)	0.296*** (0.0693)	0.277*** (0.0717)
Received wages	−0.0013 (0.0796)		−0.0649 (0.0600)	−0.0532 (0.0602)	−0.0949 (0.0609)
Financial freedom	0.0493 (0.107)			0.2110 (0.218)	0.0630 (0.233)
Human Development Index	1.218*** (0.167)				1.189** (0.514)
Constant	−0.711*** (0.132)	0.344*** (0.030)	0.364*** (0.033)	0.234* (0.134)	−0.690 (0.426)
Observations	72	72	72	72	72
Number of countries	24	24	24	24	24
R-squared	0.784	0.606	0.613	0.620	0.659
Adjusted R-squared	0.771	0.601	0.602	0.603	0.638
F test	60.90	39.69	19.85	13.60	13.40
Prob > F	0.000	0.000	0.000	0.000	0.000

The table presents the estimates of ordinary least squares (OLS) and panel data models. Model (1) represents the results of the OLS regression. Models (2) to (4) present the results of estimating the model using the panel fixed-effects model, including one variable at a time. Model (5) includes all the variables. In these models, financial capability is the dependent variable and the FinTech is the independent variable while the three other variables (Received wages, Financial freedom, and Human Development Index) are the control variables. Data are based on three waves of panel data (2014, 2017, and 2021) from the Global Index (Demirgüç-Kunt et al., 2022) and the Eurostat Databases (Eurostat, 2024) and includes 24 European countries, i.e., all countries except Luxembourg, Slovenia, and Slovak Republic. Standard errors in parentheses and clustered by country. Significance codes are as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The variables are described in Appendices 1 and 2.

Lusardi, 2020), and arguments indicating that financial freedom is important for households making sound financial decisions (Brüggen et al., 2017).

4.2. Endogeneity

When evaluating models, the key explanatory variable, FinTech, may be endogenous. Financial capability may affect FinTech because there may be a reverse causal relationship between the component variables of the FinTech index and the component variables of the dependent variable. Another concern is the presence of omitted variables or measurement errors. Such endogeneity produces biased and inconsistent parameters. Therefore, we look for methods to avoid these issues. The study employs the instrumental variable (IV). IV is a variable with exogenous variation, Z_i , that is relevant, i.e., being strongly correlated with the independent variable of FinTech, so $Cov(Z_i, \text{financial technology}_i) \neq 0$, and valid, i.e., uncorrelated with any other determinant of *Financial capability*_{*i*}, so $Cov(Z_i, \varepsilon_i) \neq 0$ (Caner & Hansen, 2004). Hence, it affects financial capability only through FinTech.

Previous studies on technology and household finance usually apply Internet-related variables such IV. For example, Gao et al. (2018) use the Internet to investigate the effect of computer penetration on farmer income. Here, we propose using broadband Internet coverage based on speed. Individuals with high-speed Internet access can conduct financial transactions using digital technologies such as electronic bill payments and direct payroll deposits (Friedline et al., 2020). The importance of high-speed Internet access stems from its strong correlation with FinTech while being unrelated to financial capability. Exogeneity ensures that high-speed Internet access is unaffected by unobserved factors influencing FinTech and financial capability, lowering the risk of endogeneity. Table 3 shows that FinTech positively and significantly affects financial capability. Model (1) includes the main variables; Model (2) represents the results after replacing the received wage with GDP per capita growth (annual %), and Model (3) presents the results after adding the additional control variables.

Table 3

Estimations for financial capability equation using instrumental variable.

VARIABLES	Model (1)	Model (2)	Model (3)
	Instrumental Variable 1	Instrumental Variable 2	Instrumental Variable 3
FinTech	0.390*** (0.0848)	0.425*** (0.1080)	0.344*** (0.0807)
Received wages	−0.160* (0.0830)		−0.136* (0.0727)
Financial freedom	−0.212 (0.282)	−0.201 (0.309)	−0.116 (0.236)
Human Development Index	1.051* (0.556)	0.747 (0.631)	1.945*** (0.607)
International openness			−0.312** (0.146)
Enterprises using software solutions			−0.0478 (0.0293)
Stock market return			0.0329 (0.0302)
GDP per capita growth (annual %)		−0.203 (0.240)	
Constant	−0.411 (0.476)	−0.243 (0.574)	−1.162** (0.496)
Observations	72	72	72
Number of countries	24	24	24

The table presents the results of the instrumental variable. Model (1) includes the main variables. Models (2) represents the results after replacing the received wage with the gross domestic product (GDP) per capita growth, annual percentage, i.e., GDP per capita growth (annual %). Model (3) presents the results after adding the additional control variable. In these models, financial capability is the dependent variable, and the FinTech is the independent variable while the three other variables (Received wages, Financial freedom, and Human Development Index) are the control variable. Data are based on three waves of panel data (2014, 2017, and 2021) from the Global Findex (Demirgüç-Kunt et al., 2022) and the Eurostat Databases (Eurostat, 2024) and includes 24 European countries, i.e., all countries except Luxembourg, Slovenia, and Slovak Republic. Significance codes are as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The variables are described in Appendices 1 and 2.

Table 4

Weak-instruments test.

Cragg–Donald Wald F statistic	50.07
Stock–Yogo weak ID test critical values	
10% Maximal IV size	16.38
15% Maximal IV size	8.96
20% Maximal IV size	6.66
25% Maximal IV size	5.53

The table presents the results of the Cragg–Donald Wald F statistic. Data is based on three waves of panel data (2014, 2017, and 2021) from the Global Findex (Demirgüç-Kunt et al., 2022) and the Eurostat Databases (Eurostat, 2024) and includes 24 European countries, i.e., all countries except Luxembourg, Slovenia, and Slovak Republic. Significance codes are as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The Cragg–Donald Wald test, which looks for weak identification,⁵ rejects the null hypothesis and shows that the endogenous regressors are significant. Table 4 shows that the F statistic (50.067) is greater than the Stock–Yogo critical values, which suggests the importance of using IV.

Another estimation strategy is the General Method of Moments (GMM), developed by Hansen (1982), Holtz-Eakin et al. (1988), Arellano and Bond (1991), Arellano and Bover (1995), and Blundell and Bond (1998). GMM is based on the concept of instrumentation, which was developed to deal with the likely correlation between independent variables and error terms, which is the most important source of endogeneity.

Holtz-Eakin et al. (1988) and Arellano and Bond (1991) developed a Difference-GMM estimator for dynamic panel data regressions that used the regressors' lagged levels as instruments for the equation in first differences. However, because of the short time series in our study, this estimator suffers from small-sample bias, making the lagged levels poor instruments for the subsequent first differences. Furthermore, using Difference-GMM results in downwardly biased standard errors when the error terms are not homogeneous. This estimator is known to be somewhat inefficient when the instruments are weak because it only considers the information contained in

⁵ The weak-instruments problem arises when the correlations between the endogenous regressors and the excluded instruments are nonzero but small.

Table 5

Fixed Effects, System-GMM, Generalized Least squares, and Weighted Least squares.

	Model (1)	Model (2)	Model (3)	Model (4)
VARIABLES	FE4	System-GMM	GLS	WLS
FinTech	0.277*** (0.0717)	0.239*** (0.0746)	0.251*** (0.0483)	0.156*** (0.0572)
Received wages	−0.0949 (0.0609)	−0.0624 (0.0572)	−0.0660 (0.0545)	−0.3000*** (0.0593)
Financial freedom	0.0630 (0.233)	0.1040 (0.226)	0.0614 (0.121)	0.7830*** (0.0926)
Human Development Index	1.189** (0.514)	1.375*** (0.341)	1.191*** (0.190)	1.217*** (0.143)
Constant	−0.690 (0.426)	−0.876*** (0.338)	−0.690*** (0.157)	−1.014*** (0.109)
Observations	72	72	72	72
Number of countries	24	24	24	24
Number of time	3	3	3	3
R-squared	0.659		0.932	0.900
Adjusted R-squared	0.638		0.928	0.894
F test	13.40		229.9	150.4
Prob >F	0.000		0.000	0.000
Arellano-Bond AR(1), p-value		0.819		
Arellano-Bond AR(2), p-value		–		
Hansen, p-value		0.722		

The table presents the tests of fixed-effects (FE4), System-GMM, generalized least-squares (GLA), and weighted least-squares (WLS) models. Model (1) includes fixed-effects regressions that use robust standard errors clustered by country. Model (2) presents System-GMM estimation for dynamic panel data models, based on an annual unbalanced panel covering 72 cases of three periods in 24 EU countries and spanning between 2014 and 2021. Constant and time dummies are not shown for brevity. All explanatory variables are treated as endogenous. System-GMM regressions collapse the instrument matrix. Orthogonal deviation is used to transform variables. In the case of two-step GMM, the Windmeijer (2005) finite sample correction for standard errors is employed. Hansen test is used for testing over-identifying restrictions in the model. Arellano-Bond AR (1) examines the hypothesis that the error terms in first differences are not serially correlated. Arellano-Bond AR (2) is not estimated because the $T = 3$ and the existence of missing values. Model (3) represents the results of GLS model to account for serial correlation. Model (4) represents the results of WLS model to account for heteroscedasticity. Robust standard errors are in parentheses. Significance codes are as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The variables are described in Appendices 1 and 2.

Table 6

Robustness check using (GDP) per capita growth.

	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
VARIABLES	OLS	FE1	FE2	FE3	FE4
FinTech	0.214*** (0.0586)	0.311*** (0.0493)	0.266*** (0.0623)	0.250*** (0.0685)	0.237*** (0.0718)
GDP per capita growth (annual %)	−0.0849 (0.165)		0.198 (0.156)	0.154 (0.152)	0.0352 (0.144)
Financial freedom	0.0547 (0.0994)			0.168 (0.217)	0.122 (0.219)
Human Development Index	1.187*** (0.173)				0.976* (0.551)
Constant	−0.690*** (0.136)	0.344*** (0.030)	0.364*** (0.035)	0.259* (0.138)	−0.568 (0.491)
Observations	72	72	72	72	72
Number of countries	24	24	24	24	24
R-squared	0.785	0.606	0.618	0.621	0.646
Adjusted R-squared	0.772	0.601	0.606	0.605	0.625
F test	61.21	39.69	20.23	13.74	12.95
Prob > F	0.000	0.000	0.000	0.000	0.000

The table presents the results of the robustness check after replacing the received wage with the gross domestic product (GDP) per capita growth, annual percentage, i.e., GDP per capita growth (annual %), Model (1) represents the results of the OLS test. Models (2) to (4) present the results of estimating the model using the panel fixed-effects model, including one variable at a time. Model (5) includes all the variables. In these models, financial capability is the dependent variable, and the FinTech is the independent variable while the three other variables (GDP per capita growth (annual %), Financial freedom, and Human Development Index) are the control variables. Data are based on three waves of panel data (2014, 2017, and 2021) from the Global Findex (Demirgüç-Kunt et al., 2022) and the Eurostat Databases (Eurostat, 2024) and includes 24 European countries, i.e., all countries except Luxembourg, Slovenia, and Slovak Republic. Standard errors in parentheses and significance codes are as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The variables are described in Appendices 1 and 2.

the differences. [Blundell and Bond \(1998\)](#) suggest adding additional information to the differences. The System-GMM estimator's combination of moment restrictions for differences and levels results in estimates with lower bias and higher accuracy ([Bond, 2002](#)).

[Table 5](#) shows that the Hansen test fails to reject the null hypothesis, implying that the over-identifying restrictions are valid. Similarly, Arellano-Bond AR (1) does not reject the null hypothesis, confirming that the error terms in the first differences are not serially correlated. Model (2) System-GMM confirms the positive significant effect of FinTech on financial capability at the 1% level, as reported in Model (1) FE4, with a relatively similar magnitude. In a similar vein, Model (3) GLS and Model (4) WLS, which represent the generalized least squares ([Kariya & Kurata, 2004](#)) and weighted least-squares results ([Xu & Eckstein, 1995](#)), respectively, confirm the positive and significant effect of FinTech.

4.3. Robustness check

Running the pooled OLS and fixed-effects models after replacing the received wage variable with GDP per capita growth (annual %) provides a robustness check. [Table 6](#) shows significant results for all five models, which are similar to the baseline model in [Table 2](#). It is also visible that the coefficients do not change significantly.

The analysis also includes four additional variables: rule of law, international openness, enterprises using software solutions, and stock market return. The findings in [Table 7](#) confirm the significant effect of FinTech on financial capability. All FinTech coefficients are significant at the 1% level, with magnitudes ranging from 0.251 to 0.328, whereas trade openness (i.e., international openness) and technology utilization in the business sector (i.e., enterprises using software solutions) are significant at the 5% level when inserted separately in Models (2) and (3). The stock market's return is significant at 10%, as shown in Model (4). Model (5) includes all variables and concludes that international openness has a significant negative effect on financial capability at the 10% level, supporting the conclusion of [Acar and Dogruel \(2012\)](#) that such openness reduces income inequality.

The EU includes countries with varying levels of economic and institutional development. To mitigate the negative effects of this heterogeneity, the EU implemented a growth strategy for 2011–2020. The Europe 2020 strategy focused on five ambitious goals: employment, innovation, education, social inclusion, and climate/energy. To achieve these objectives, each country must establish its own national targets ([European Union, 2014](#)).

We divide the EU countries involved in our investigation into two subsamples and rerun the regression on them. To divide the sample, we used the 2015 ranking of EU countries based on their level of implementation of the 2020 Europe strategy objectives ([Fura et al., 2017](#)). Countries making good progress toward implementing the Europe 2020 strategy and score above the median represent the first subsample, while countries that do not meet this threshold represent the second subsample. [Table 8](#) shows the results of the subsample tests. Models (1) and (2) represent the results of countries with higher levels of strategy implementation, whereas Models (3) and (4) represent countries with lower levels. The four models assert that FinTech has a significant effect on financial capability. Notably, the effect on FinTech is larger (and statistically significant at the 1% level) in countries that are well on their way to implementing the strategy than in the other countries.

5. Conclusion

5.1. Policy recommendations

This study uses three waves of Global Findex Panel Data (2014, 2017, and 2021) and the Eurostat database to assess financial capability in 24 of the EU's 27 countries and investigate the role of FinTech in addressing financial capability. The results show a strong correlation between FinTech and financial capability. The Human Development Index positively and significantly affects financial capability, while received wages and financial freedom have no effect.

The positive and significant impact of FinTech on financial capability highlights the role of financial service providers in enhancing individual financial capability. Those who use FinTech solutions receive newsletters and other forms of communication containing useful financial information via their devices. These findings echo the results of [Karlan et al. \(2016\)](#), who argue that reminder messages sent by banks improve customer commitment to saving goals. Furthermore, FinTech solutions' opportunities for individuals to control and trace their financial transactions may allow them to conduct sound financial planning ([Nourallah & Öhman, 2021](#)).

The study also suggests that the higher the achievements of European countries in terms of human development, the greater households' potential financial capability. In terms of income, an increase in the proportion of people receiving wages appears to have no effect on financial ability. This contradicts the findings of [Almenberg et al. \(2021\)](#) and [Lusardi et al. \(2017\)](#), but it supports the argument that higher income does not guarantee a stable financial position. For example, [Atkinson et al. \(2007\)](#) report that low-income households with good money management may be better financially than higher-wage households with poor money management skills. This study also found that financial freedom has a limited impact on people's financial abilities. This contradicts the suggested role of financial freedom in banks ([Chortareas et al., 2013](#)), implying that financial freedom can serve an important function in the banking sector but may have a limited role in individual financial capability.

Using Sen's seminal capability approach ([Sen, 1993; 2005](#)), this study identifies five key variables for measuring financial ability:

skills, debt, savings, financial resilience, and financial well-being. Implementing these variables reveals some issues related to different levels of financial capability across the EU. First, the countries studied have different financial capability scores. Some countries, including Sweden, have high scores, while others, such as Bulgaria, have low scores. Second, a descriptive analysis of growth rates revealed that Hungary and Latvia made significant progress in increasing financial capability between 2014 and 2021; however, these countries still need to do much more before reaching an acceptable level. It is also worth noting that a country with a relatively high financial capability score, such as Denmark, experienced negative growth over the studied period.

When it comes to individual use of FinTech solutions, the analysis shows that the Nordic–Baltic region, particularly Sweden, Finland, and Estonia, performs exceptionally well. Despite their low FinTech scores, Romania and Bulgaria saw significant increases in FinTech usage between 2014 and 2021.

EU policymakers must consider financial issues in future planning. Factors that enable countries like Sweden to achieve high financial capability and FinTech scores are suggested for further investigation, and “lessons to learn” should be shared with other countries. In contrast to North America—see the US national financial capability study (FINRA Investor Education Foundation, 2023) and the Canadian financial capability survey (Government of Canada, 2023)—there appears to be a lack of comprehensive scales capturing financial capability in the EU. Based on the findings of this study, it appears that developing a financial capability section in the Household Finance and Consumption Survey (European Central Bank, 2023) that includes financial well-being variables would be beneficial. Furthermore, the same suggestion could improve national surveys across the EU. Examples include the German Panel on Household Finances, which examines German households’ balance sheets, pensions, income, work life, and demographic variables (German Federal Bank, 2024), and the DNB Household Survey in the Netherlands (Institute for Data Collection and Research, 2023), which focuses on the psychological and economic aspects of financial behavior (Bruijne & Streefkerk, 2020).

Policy debates in the EU have focused on increasing public financial literacy, particularly in light of the COVID-19 pandemic and the conflict in Ukraine. However, we must remember that financial literacy is a passive approach. In addition to efforts to improve financial literacy, a more proactive approach should be considered. In this sense, financial capability appears appropriate.

Table 7
Robustness check using additional control variables.

VARIABLES	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)
FinTech	0.272*** (0.0740)	0.328*** (0.0728)	0.263*** (0.0638)	0.251*** (0.0754)	0.288*** (0.0713)
Received wages	−0.0969 (0.0621)	−0.1220 (0.0732)	−0.0878 (0.0614)	−0.0885 (0.0654)	−0.1050 (0.0702)
Financial freedom	0.0520 (0.239)	0.0509 (0.243)	−0.0026 (0.203)	0.0261 (0.249)	−0.0075 (0.218)
Human Development Index	1.222** (0.519)	2.061*** (0.646)	1.649*** (0.522)	0.859 (0.511)	1.881*** (0.610)
Rule of law	−0.0472 (0.0772)				
International openness		−0.394** (0.159)			−0.262* (0.150)
Enterprises using software solutions			−0.0800** (0.0293)		−0.0513 (0.0360)
Stock market return				0.0721* (0.0386)	0.0398 (0.0358)
Constant	−0.665 (0.430)	−1.377** (0.537)	−1.008** (0.388)	−0.370 (0.464)	−1.174** (0.513)
Observations	72	72	72	72	72
Number of countries	24	24	24	24	24
R-squared	0.660	0.718	0.712	0.701	0.756
Adjusted R-squared	0.634	0.696	0.690	0.679	0.729
F test	12.86	22.58	18.08	13.13	19.65
Prob > F	0.000	0.000	0.000	0.000	0.000

The table presents the fixed-effects results of the robustness check after adding additional control variables (Rule of law, International openness, Enterprises using software solutions, and Stock market return). In these models, financial capability is the dependent variable, and the FinTech is the independent variable while the three other variables (Received wages, Financial freedom, Human Development Index), and other variables are the additional control variables. Model (1) represents the results of including rule of law, Model (2) represents the results of including International openness, Model (3) represents the results of including Enterprises using software solutions, Model (4) represents the results of including stock market return, Model (5) represents the results of including all variables. Data are based on three waves of panel data (2014, 2017, and 2021) from the Global Findex (Demirgüç-Kunt et al., 2022) and the Eurostat Databases (Eurostat, 2024) and includes 24 European countries, i.e., all countries except Luxembourg, Slovenia, and Slovak Republic. Standard errors in parentheses and clustered by country. Significance codes are as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The variables are described in Appendices 1 and 2.

Table 8
Robustness check using subsamples.

	Model (1)	Model (2)	Model (3)	Model (4)
VARIABLES	Subsample toward EU2020 Strategy	Subsample toward EU2020 Strategy	Subsample far away from EU2020 Strategy	Subsample far away from EU2020 Strategy
FinTech	0.358*** (0.0922)	0.377*** (0.100)	0.267** (0.0935)	0.325*** (0.0883)
Received wages	−0.0471 (0.0777)	−0.0851 (0.100)	−0.1230* (0.0681)	−0.2010** (0.0805)
Financial freedom	0.407 (0.362)	0.0593 (0.328)	−0.0170 (0.342)	−0.133 (0.312)
Human Development Index	−0.666 (0.882)	−0.442 (1.085)	1.910** (0.784)	2.794*** (0.656)
International openness		−0.261* (0.123)		−0.231 (0.195)
Enterprises using software solutions		0.0482 (0.0342)		−0.106** (0.0421)
Stock market return		0.0752** (0.0260)		0.00686 (0.0650)
Constant	0.679 (0.611)	0.754 (0.817)	−1.248* (0.618)	−1.832*** (0.491)
Observations	36	36	36	36
Number of countries	12	12	12	12
R-squared	0.685	0.793	0.700	0.826
Adjusted R-squared	0.644	0.741	0.661	0.782
F test	11.97	13.41	8.129	14.58
Prob >F	0.000	0.000138	0.003	0.000

The Table presents the results of using subsample tests. The sample is divided into two subsamples. In these models, financial capability is the dependent variable, and the FinTech is the independent variable while the three other variables (Received wages, Financial freedom, Human Development Index), and other variables are the additional control variables (International openness, Enterprises using software solutions, and Stock market return). Models (1) and (2) represent the results of the countries that progress well toward the Europe 2020 strategy. While Models (3) and (4) represent the countries with a humble level of implementation of the Europe 2020 strategy. Data are based on three waves of panel data (2014, 2017, and 2021) from the Global Findex (Demirgüç-Kunt et al., 2022) and the Eurostat Databases (Eurostat, 2024) and includes 24 European countries, i.e., all countries except Luxembourg, Slovenia, and Slovak Republic. All the models represent the results of estimating the model using the panel fixed-effects model. Standard errors in parentheses are clustered by country. Significance codes are as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The variables are described in Appendices 1 and 2.

5.2. Limitations and suggestions for further research

Even though income and financial freedom can help improve financial issues (Collins & Urban, 2020; Lusardi et al., 2017), other factors appear to be at least as important. To address these issues, alternative methodologies such as surveys and qualitative studies can be used. This study does not explore the role of socioeconomic variables because of a lack of data. Therefore, we recommend future studies should address these factors. The three waves of data used here have some limitations, and future studies could try to access other datasets covering longer periods.

According to the present results, the proliferation of foreign products and services and the use of big data analytics to design marketing campaigns appear to mitigate household financial capability. Future studies could explore the potential impact of marketing campaign exposure on financial capability.

FinTech is possibly associated with some unethical financial behavior, such as overconsumption and fraud (Gomber et al., 2018). However, proper use of FinTech can have positive consequences, not least in terms of increasing financial capability. This study can be expanded upon to explore what factors can improve the proper use of FinTech solutions. Future studies may also focus on financial capability in particular groups of people, such as the young generation. This generation lacks financial experience but possesses strong technological skills (Nourallah & Öhman, 2021), so it may be of interest to investigate how to use FinTech solutions to improve financial capability among this group of individuals, which would have a significant impact on how the society of the future will be shaped.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Mustafa Nourallah: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Peter Öhman:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Conceptualization. **Samer Hamati:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no conflict of interest.

Data availability

Data will be made available on request.

Acknowledgments

The authors would like to acknowledge InfraVis for providing application expertise for visualization through Swedish Research Council grant 2021-00181.

Appendix A. Appendix

Appendix 1

Definition of variables used to measure financial capability and financial technology.

Constructs	Variables	Definitions	Datasets
Skills	Participation rate in education and training (age 18 to 64)	“participation in formal and non-formal education and training. In this section the reference period for the participation in education and training is the four weeks prior to the interview. Participation rates in education and training for various age groups and by different breakdowns are presented.” (Eurostat, 2024)	Eurostat
Debt	Borrowed for health or medical purposes (% age 15+)	“the percentage of respondents who report borrowing any money for health or medical purposes in the past year” (Demirgüç-Kunt et al., 2022).	The Global Findex Database

(continued on next page)

Appendix 1 (continued)

Constructs	Variables	Definitions	Datasets
Saving	Borrowed from family or friends (% age 15+)	“the percentage of respondents who report borrowing any money from family, relatives, or friends in the past year” (Demirgüç-Kunt et al., 2022).	
	Saved any money (% age 15+)	“the percentage of respondents who report personally saving or setting aside any money for any reason and using any mode of saving in the past year” (Demirgüç-Kunt et al., 2022).	
	Saved for old age (% age 15+)	“the percentage of respondents who report saving or setting aside any money in the past year for old age” (Demirgüç-Kunt et al., 2022).	
	Saved at a financial institution (% age 15+)	“the percentage of respondents who report saving or setting aside any money at a bank or another type of financial institution in the past year” (Demirgüç-Kunt et al., 2022).	
Financial resilience	Ability to face unexpected financial expenses	“capacity to face unexpected expenses” (Eurostat, 2021).	Eurostat
Financial well-being	Average rating of satisfaction	overall life satisfaction (Eurostat, 2024) ^a	
	Distribution of population aged 18 and over by health status very good	distribution of population aged 18 to 64 years who answered that their health status is very good (Eurostat, 2024)	
Financial technology	Received digital payments (% age 15+)	“the percentage of respondents who report using a mobile money account, a debit or credit card, or a mobile phone to receive a payment into an account in the past year. This includes respondents who report receiving remittances, receiving payments for agricultural products, receiving government transfers, receiving wages, or receiving a public sector pension directly into a financial institution account or into a mobile money account in the past year” (Demirgüç-Kunt et al., 2022).	The Global Findex Database
	made a digital payment (% age 15+)	“the percentage of respondents who report using mobile money, a debit or credit card, or a mobile phone to make a payment from an account; or who report using the internet to pay bills or to buy something online or in a store in the past year. This includes respondents who report paying bills or sending remittances directly from a financial institution account or through a mobile money account in the past year” (Demirgüç-Kunt et al., 2022).	
	made a utility payment using a mobile phone (% age 15+)	“the percentage of respondents who report personally making regular payments for water, electricity, or trash collection in the past year using a mobile phone” (Demirgüç-Kunt et al., 2022).	
	Individuals' frequency of internet use	“frequency of internet access: once a week (including every day).” (Eurostat, 2024)	

The Table presents the definitions of the variables used to measure financial capability and financial technology.

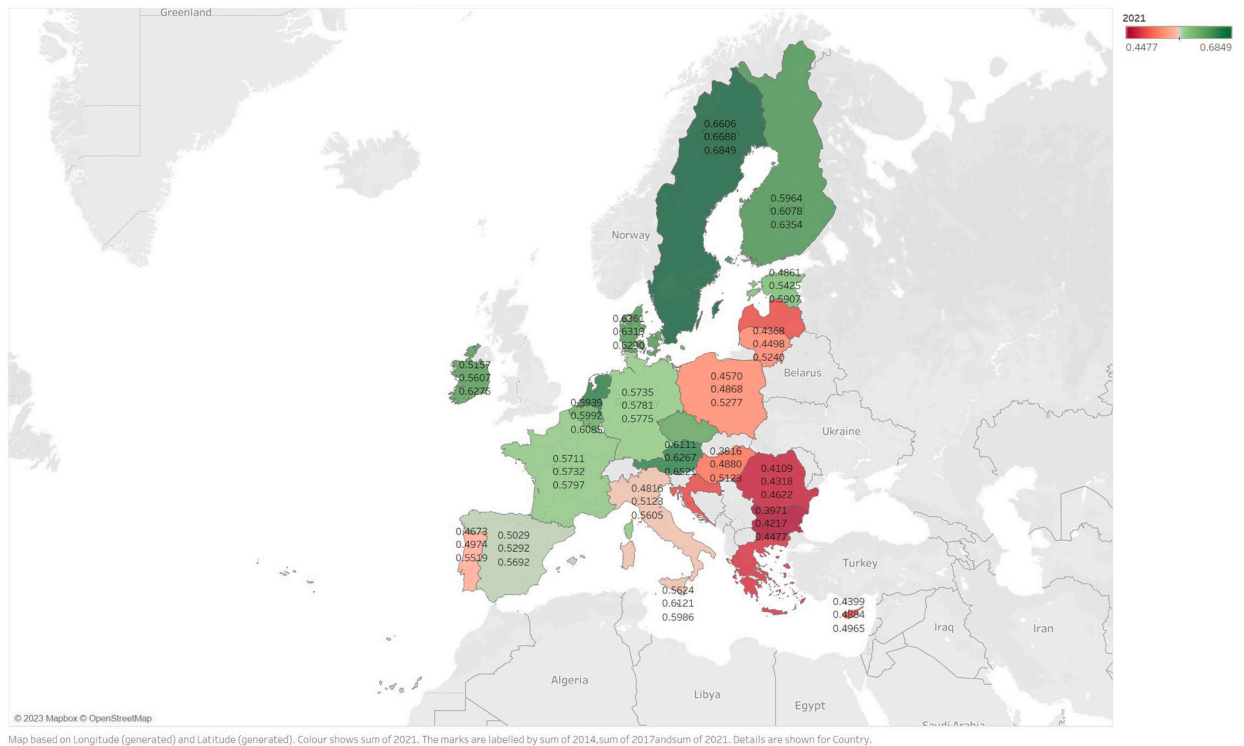
^a For more information: https://ec.europa.eu/eurostat/databrowser/view/ilc_pw01/default/table?lang=en

Appendix 2

Definition of control variables.

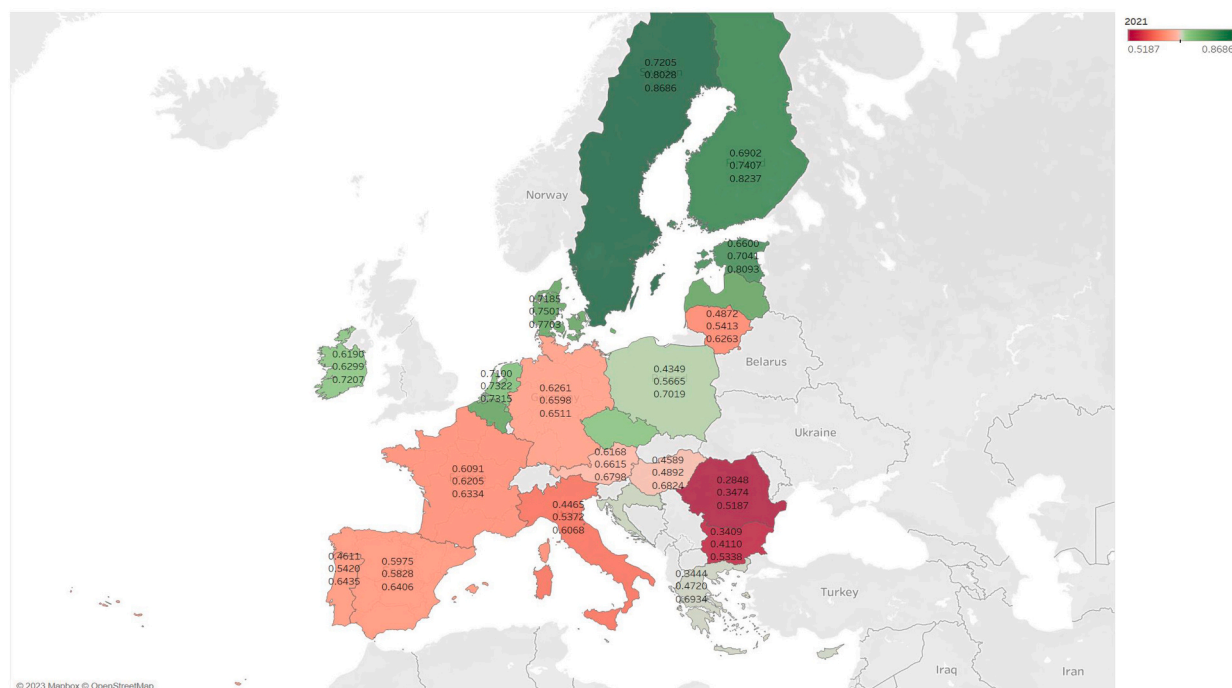
Constructs	Definitions	Datasets
Received wages (% age 15+)	“the percentage of respondents who report receiving any money from an employer in the past year in the form of a salary or wages for doing work. This does not include any money received directly from clients or customers” (Demirgüç-Kunt et al., 2022).	The Global Findex Database
Financial freedom	“is the fundamental right of every human to control his or her own labour and property.” (Heritage Foundation, 2022b)	Heritage Foundation
Human Development Index	“is a summary measure of average achievement in key dimensions of human development: a long and healthy life, being knowledgeable and have a decent standard of living” (United Nations Development Programme, 2022).	Human Development Index
The Gross Domestic Product (GDP) per capita growth, annual percentage, i.e., GDP per capita growth (annual %)	“Annual percentage growth rate of GDP per capita based on constant local currency. GDP per capita is gross domestic product divided by midyear population. GDP at purchaser's prices is the sum of gross value added by all resident producers in the economy plus any product taxes and minus any subsidies not included in the value of the products. It is calculated without making deductions for depreciation of fabricated assets or for depletion and degradation of natural resources” (The World Bank, 2023).	The World Bank
Rule of law	“law is a durable system of laws, institutions, norms, and community commitment that delivers four universal principles: accountability, just law, open government, and accessible and impartial justice” (World justice project, 2021)	World Justice Project
International openness	“the sum of exports and imports of goods and services measured as a share of gross domestic product” (The World Bank, 2023)	The World Bank
Enterprises using software solutions	“Enterprises using software solutions, like CRM to analyse information about clients for marketing purposes” (Eurostat, 2024)	(Eurostat, 2024)
Stock market return	“Stock market return is the growth rate of annual average stock market index” (The World Bank, 2023).	The World Bank

The Table presents the definitions of the control variables.



Appendix 3. Financial capability score.

Note: Appendix 3 visualizes the score financial capability in the 24 EU countries under study. Color details based on 2021 and countries were classified as either high in financial capability (assigned in green color) or low in financial capability (assigned in red color). Each of these countries indicates three numbers, where the bottom number represents 2021, the number in the middle is 2017, and the top number is 2014. Data is based on three waves of panel data (2014, 2017, and 2021) of the Global Findex and the Eurostat Databases. Maps generated by Tableau.



Appendix 4. Financial technology score.

Note: Appendix 4 visualizes the score of FinTech in the 24 EU countries under study. Color details based on 2021 and countries were classified as either high in FinTech (assigned in green color) or low in FinTech (assigned in red color). Each of these countries indicates three numbers, where the bottom number represents 2021, the number in the middle is 2017, and the top number is 2014. Data is based on three waves of panel data (2014, 2017, and 2021) of the Global Findex and the Eurostat Databases. Maps generated by Tableau.

References

- Acar, S., & Dogruel, F. (2012). Sources of inequality in selected MENA countries. *Structural Change and Economic Dynamics*, 23, 276–285. <https://doi.org/10.1016/j.strueco.2012.03.004>
- Aixalá, J., & Fabro, G. (2008). Does the impact of institutional quality on economic growth depend on initial income level? *Economic Affairs*, 28, 45–49. <https://doi.org/10.1111/j.1468-0270.2008.00844.x>
- Almenberg, J., Lusardi, A., Sävje-Söderbergh, J., & Vestman, R. (2021). Attitudes towards debt and debt behavior. *The Scandinavian Journal of Economics*, 123, 780–809. <https://doi.org/10.1111/sjoe.12419>
- Alt, R., Beck, R., & Smits, M. T. (2018). FinTech and the transformation of the financial industry. *Electronic Markets*, 28, 235–243. <https://doi.org/10.1007/s12525-018-0310-9>
- Arellano, M., & Bond, S. (1991). Some tests of specification for panel data: Monte Carlo evidence and an application to employment equations. *Review of Economic Studies*, 58, 277–297. <https://doi.org/10.2307/2297968>
- Arellano, M., & Bover, O. (1995). Another look at the instrumental variable estimation of error-components models. *Journal of Econometrics*, 68, 29–51. [https://doi.org/10.1016/0304-4076\(94\)01642-D](https://doi.org/10.1016/0304-4076(94)01642-D)
- Atkinson, A., McKay, S., Collard, S., & Kempson, E. (2007). Levels of financial capability in the UK. *Public Money & Management*, 27, 29–36. <https://doi.org/10.1111/j.1467-9302.2007.00552.x>
- Baltagi, B., & Baltagi, B. (2008). *Econometric Analysis of Panel Data* (4th ed.). Chichester, UK: Wiley.
- Barrafrem, K., Västfjäll, D., & Tinghög, G. (2020). Financial well-being, COVID-19, and the financial better-than-average effect. *Journal of Behavioral and Experimental Finance*, 28, Article 100410. <https://doi.org/10.1016/j.jbef.2020.100410> (PubMed: 33042778).
- Birkenmaier, J., Maynard, B., & Kim, Y. (2022). Interventions designed to improve financial capability: A systematic review. *Campbell Systematic Reviews*, 18, Article e1225. <https://doi.org/10.1002/CL2.1225> (PubMed: 36913196).
- Birkenmaier, J., Rothwell, D., & Agar, M. (2022). How is consumer financial capability measured? *Journal of Family and Economic Issues*, 43, 654–666. <https://doi.org/10.1007/s10834-022-09825-4> (PubMed: 36373053).
- Blundell, R., & Bond, S. (1998). Initial conditions and moment restrictions in dynamic panel data models. *Journal of Econometrics*, 87, 115–143. [https://doi.org/10.1016/S0304-4076\(98\)00009-8](https://doi.org/10.1016/S0304-4076(98)00009-8)
- Bon, V. (2021). The digitalization–economic growth relationship in developing countries and the role of governance. *Scientific Annals of Economics and Business*, 68, 481–493. <https://doi.org/10.47743/saeb-2021-0028>
- Bond, S. R. (2002). Dynamic panel data models: A guide to micro data methods and practice. *Portuguese Economic Journal*, 1, 141–162. <https://doi.org/10.1007/s10258-002-0009-9>
- Brounen, D., Koedijk, K. G., & Pownall, R. A. J. (2016). Household financial planning and savings behavior. *Journal of International Money and Finance*, 69, 95–107. <https://doi.org/10.1016/j.jimonfin.2016.06.011>
- Browning, M., & Lusardi, A. (1996). Household saving: Micro theories and micro facts. *Journal of Economic Literature*, 34, 1797–1855.

- Brüggen, E. C., Hogreve, J., Holmlund, M., Kabadayi, S., & Löfgren, M. (2017). Financial well-being: A conceptualization and research agenda. *Journal of Business Research*, 79, 228–237. <https://doi.org/10.1016/j.jbusres.2017.03.013>
- Bruijne, M. D., & Streefkerk, M. (2020). *No Data Management and Preservation Policy of the DNB Household Survey (DHS)*.
- Caner, M., & Hansen, B. E. (2004). Instrumental variable estimation of a threshold model. *Econometric Theory*, 20, 813–843. <https://doi.org/10.1017/S0266466604205011>
- Chortareas, G. E., Girardone, C., & Ventouri, A. (2013). Financial freedom and bank efficiency: Evidence from the European Union. *Journal of Banking and Finance*, 37, 1223–1231. <https://doi.org/10.1016/j.jbankfin.2012.11.015>
- Collins, J. M., & Urban, C. (2020). Measuring financial well-being over the lifecycle. *European Journal of Finance*, 26, 341–359. <https://doi.org/10.1080/1351847X.2019.1682631>
- D'Acunto, F., & Rossi, A. G. (2023). Robo-advice: Transforming households into rational economic agents. *Annual Review of Financial Economics*, 15, 543–563. <https://doi.org/10.1146/annurev-financial-110921-013217>
- Demir, A., Pesqué-Cela, V., Altunbas, Y., & Murinde, V. (2022). Fintech, financial inclusion and income inequality: A quantile regression approach. *European Journal of Finance*, 28, 86–107. <https://doi.org/10.1080/1351847X.2020.1772335>
- Demirgüç-Kunt, A., Klapper, L., Singer, D., & Ansar, S. (2022). *The Global Findex Database 2021*. World Bank. <https://doi.org/10.1596/978-1-4648-1897-4>
- European Banking Authority. (2024). *Financial Innovation and FinTech*.
- European Central Bank. (2023). Household Finance and Consumption Survey (HFCS). Retrieved from https://www.ecb.europa.eu/stats/ecb_surveys/hfcs/html/index.en.html.
- European Central Bank. (2024). Digital Euro. Retrieved from https://www.ecb.europa.eu/euro/digital_euro/html/index.en.html.
- European Union-Organization for Economic Co-operation and Development. (2022). Financial Competence Framework for Adults in the European Union. Retrieved from <https://www.oecd.org/finance/financial-competence-framework-for-adults-in-the-European-Union.htm>.
- Eurostat. (2021). Glossary: Severe Material and Social Deprivation Rate (SMSD)—Statistics Explained. [https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Severe_material_and_social_deprivation_rate_\(SMSD\)](https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Severe_material_and_social_deprivation_rate_(SMSD)).
- Eurostat. (2024). Eurostat-Database. <https://ec.europa.eu/eurostat/web/main/data/database>.
- Financial Consumer Agency of Canada. (2019). Data Collection for the 2019 Canadian Financial Capability Survey. https://publications.gc.ca/collections/collection_2019/acfc-fcac/FC5-42-2-2019-1-eng.pdf.
- FINRA Investor Education Foundation. (2023). National Financial Capability Study. <https://finrafoundation.org/knowledge-we-gain-share/nfcs/about-nfcs>.
- French, D., McKillop, D., & Stewart, E. (2020). The effectiveness of smartphone apps in improving financial capability. *European Journal of Finance*, 26, 302–318. <https://doi.org/10.1080/1351847X.2019.1639526>
- Friedline, T., Narahariseti, S., & Weaver, A. (2020). Digital redlining: Poor rural communities' access to Fintech and implications for financial inclusion. *Journal of Poverty*, 24, 517–541. <https://doi.org/10.1080/10875549.2019.1695162>
- Fura, B., Wojnar, J., & Kasprzyk, B. (2017). Ranking and classification of EU countries regarding their levels of implementation of the Europe 2020 strategy. *Journal of Cleaner Production*, 165, 968–979. <https://doi.org/10.1016/j.jclepro.2017.07.088>
- Gao, Y., Zang, L., & Sun, J. (2018). Does computer penetration increase farmers' income? An empirical study from China. *Telecommunications Policy*, 42, 345–360. <https://doi.org/10.1016/j.TELPOL.2018.03.002>
- German Federal Bank. (2024). About the PHF. Retrieved from <https://www.bundesbank.de/en/bundesbank/research/panel-on-household-finances/about-the-phf/about-the-phf-617320>.
- Gomber, P., Kauffman, R. J., Parker, C., & Weber, B. W. (2018). On the Fintech revolution: Interpreting the forces of innovation, disruption, and transformation in financial services. *Journal of Management Information Systems*, 35, 220–265. <https://doi.org/10.1080/07421222.2018.1440766>
- Government of Canada. (2023). Canadians and Their Money: Key Findings From the 2019 Canadian Financial Capability Survey. Retrieved from <https://www.canada.ca/en/financial-consumer-agency/programs/research/canadian-financial-capability-survey-2019.html>.
- Greene, W. (2008). *Econometric Analysis* (6th ed.). Prentice Hall.
- Hansen, L. P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica*, 50, 1029–1054. <https://doi.org/10.2307/1912775>
- Hastings, J. S., Madrian, B. C., & Skimmyhorn, W. L. (2013). Financial literacy, financial education, and economic outcomes. *Annual Review of Economics*, 5, 347–373. <https://doi.org/10.1146/annurev-economics-082312-125807>
- Hausman, J. A. (1978). Specification tests in econometrics. *Econometrica*, 46, 1251. <https://doi.org/10.2307/1913827>
- Heritage Foundation. (2022a). Index of Economic Freedom [Methodology]. Retrieved from https://www.heritage.org/index/pdf/2022/book/02_2022_indexofeconomicfreedom_methodology.pdf.
- Heritage Foundation. (2022b). Index of Economic Freedom. Retrieved from <https://www.heritage.org/index/about>.
- Hoffmann, A., & Otteby, K. (2018). Personal finance blogs: Helpful tool for consumers with low financial literacy or preaching to the choir? *International Journal of Consumer Studies*, 42, 241–254. <https://doi.org/10.1111/ijcs.12412>
- Holtz-Eakin, D., Newey, W., & Rosen, H. S. (1988). Estimating vector autoregressions with panel data. *Econometrica*, 56, 1371–1395. <https://doi.org/10.2307/1913103>
- Iddrisu, A. G., & Chen, B. (2022). Economic growth through digitalization in Africa: Does financial sector development play a mediating role? *International Journal of Emerging Markets*. <https://doi.org/10.1108/IJOEM-02-2022-0278>
- Institute for Data Collection and Research. (2023). DHS Data Access. Retrieved from <https://www.dhsdata.nl/site/users/login>.
- Johnson, E., & Sherraden, M. S. (2007). From financial literacy to financial capability among youth center for social development. *Journal of Sociology and Social Welfare*, 34, 119–145. <https://doi.org/10.15453/0191-5096.3276>
- Kariya, T., & Kurata, H. (2004). *Generalized Least Squares*. Chichester, UK: John Wiley & Sons.
- Karlan, D., McConnell, M., Mullainathan, S., & Zinman, J. (2016). Getting to the top of mind: How reminders increase saving. *Management Science*, 62, 3393–3411. <https://doi.org/10.1287/mnsc.2015.2296>
- Keister, T., & Sanches, D. (2023). Should central banks issue digital currency? *Review of Economic Studies*, 90, 404–431. <https://doi.org/10.1093/restud/rdac017>
- Klapper, L., & Lusardi, A. (2020). Financial literacy and financial resilience: Evidence from around the world. *Financial Management*, 49, 589–614. <https://doi.org/10.1111/fima.12283>
- Kuvvet, E. (2022). RobinHood investors and corporate misconduct. *Global Finance Journal*, 54, Article 100752. <https://doi.org/10.1016/j.gfj.2022.100752>
- Lee, C. C., Ni, W., & Zhang, X. (2023). FinTech development and commercial bank efficiency in China. *Global Finance Journal*, 57, 1044–1283. <https://doi.org/10.1016/j.gfj.2023.100850>
- Levin, A., Lin, C. F., & Chu, C. S. (2002). Unit root tests in panel data: Asymptotic and finite-sample properties. *Journal of Econometrics*, 108, 1–24. [https://doi.org/10.1016/S0304-4076\(01\)00098-7](https://doi.org/10.1016/S0304-4076(01)00098-7)
- Lusardi, A. (2011). Americans' Financial Capability. NBER Working Papers Series. Retrieved from <http://www.nber.org/papers/w17103>.
- Lusardi, A., Hasler, A., & Yakoboski, P. J. (2021). Building up financial literacy and financial resilience. *Mind & Society*, 20, 181–187. <https://doi.org/10.1007/s11299-020-00246-0>
- Lusardi, A., Samek, A., Kapteyn, A., Glinert, L., Hung, A., & Heinberg, A. (2017). Visual tools and narratives: New ways to improve financial literacy. *Journal of Pension Economics and Finance*, 16, 297–323. <https://doi.org/10.1017/S1474747215000323>
- Nguyen, D. K., Serpinis, G., & Stasinakis, C. (2023). Big data, artificial intelligence and machine learning: A transformative symbiosis in favour of financial technology. *European Financial Management*, 29, 517–548. <https://doi.org/10.1111/eufm.12365>
- Nourallah, M. (2003). *A Tale of Two Fintech Solutions: Loyalty in Mobile Bank Applications and Initial Trust in Financial Robo-Advisors* [Doctoral Dissertation]. Mid Sweden University
- Nourallah, M., & Öhman, P. (2021). Impact of advanced technologies on consumer finance and retail investment. In R. Ho, A. H. H. Ng, & M. Nourallah (Eds.), *Impact of Globalization and Advanced Technologies on Online Business Models* (pp. 1–15). IGI Global. <https://doi.org/10.4018/978-1-7998-7603-8.ch001>.

- Nussbaum, M. (2011). *Creating Capabilities the Human Development Approach*, Cambridge, MA. Harvard University Press.
- Nuzzo, G., & Piermattei, S. (2020). Discussing measures of financial inclusion for the main euro area countries. *Social Indicators Research*, 148, 765–786. <https://doi.org/10.1007/s11205-019-02223-8>
- Ofosu-Mensah Ababio, J., Attah-Botchwey, E., Osei-Assibey, E., & Barnor, C. (2021). Financial inclusion and human development in frontier countries. *International Journal of Finance and Economics*, 26, 42–59. <https://doi.org/10.1002/IJFE.1775>
- Organization for Economic Co-operation and Development. (2023). *Household Debt*. OECD Data. <https://doi.org/10.1787/f03b6469-en>
- Panos, G. A., & Wilson, J. O. S. (2020). Financial literacy and responsible finance in the FinTech era: Capabilities and challenges. *European Journal of Finance*, 26, 297–301. <https://doi.org/10.1080/1351847X.2020.1717569>
- Sánchez, A., & Ruiz-Martos, M. J. (2018). Europe 2020 strategy and citizens' life satisfaction. *Journal of Happiness Studies*, 19, 2315–2338. <https://doi.org/10.1007/s10902-017-9928-0>
- Sen, A. (1993). In M. Nussbaum (Ed.), *The Quality of Life* (pp. 30–53). Oxford: Oxford University Press. <https://doi.org/10.1093/0198287976.001.0001>.
- Sen, A. (2005). Human rights and capabilities. *Journal of Human Development*, 6, 151–166. <https://doi.org/10.1080/14649880500120491>
- Sha'ban, M., Girardone, C., & Sarkisyan, A. (2020). Cross-country variation in financial inclusion: A global perspective. *European Journal of Finance*, 26, 319–340. <https://doi.org/10.1080/1351847X.2019.1686709>
- Sherraden, M. S., Huang, J., Jones, J. L., & Callahan, C. (2022). Building financial capability and assets in America's families. *Families in Society*, 103, 3–6. <https://doi.org/10.1177/10443894211066464>
- Statista. (2024). FinTech. Europe. Retrieved from <https://www.statista.com/outlook/dmo/fintech/europe>.
- Tahir, M. S., Shahid, A. U., & Richards, D. W. (2022). The role of impulsivity and financial satisfaction in a moderated mediation model of consumer financial resilience and life satisfaction. *International Journal of Bank Marketing*, 40, 773–790. <https://doi.org/10.1108/IJBM-09-2021-0407>
- Taylor, M. P., Jenkins, S. P., & Sacker, A. (2011). Financial capability and psychological health. *Journal of Economic Psychology*, 32, 710–723. <https://doi.org/10.1016/J.JOEP.2011.05.006>
- The World Bank. (2013). Making Sense of Financial Capability Surveys Around the World. Retrieved from <http://www.worldbank.org>.
- The World Bank. (2022). World Bank Open Data. Retrieved from <https://data.worldbank.org/>.
- The World Bank. (2023). *GDP Per Capita Growth (Annual %)*.
- Tufano, P. (2009). Consumer finance. *Annual Review of Financial Economics*, 1, 227–247. <https://doi.org/10.1146/annurev.financial.050808.114457>
- Union, E. (2014). *Europe 2020 Strategy*. Retrieved from. <https://doi.org/10.2864/52384>.
- United Nations Development Programme. (2022). *Human Development Index*. *Human Development Reports*. Retrieved from. <https://hdr.undp.org/data-center/human-development-index#/indicies/hdi>.
- Van Rooij, M., Lusardi, A., & Alessie, R. (2011). Financial literacy and stock market participation. *Journal of Financial Economics*, 101, 449–472. <https://doi.org/10.1016/j.jfineco.2011.03.006>
- Windmeijer, F. (2005). A finite sample correction for the variance of linear efficient two-step GMM estimators. *Journal of Econometrics*, 126(1), 25–51.
- Wooldridge, J. (2022). *Econometric Analysis of Cross Section and Panel Data*, Cambridge, MA. MIT Press.
- World Justice Project. (2021). Rule Of Law Index. <https://worldjusticeproject.org/sites/default/files/documents/WJP-INDEX-2021.pdf>.
- Xiao, J. J., & Bialowolski, P. (2023). Consumer financial capability and quality of life: A global perspective. *Applied Research in Quality of Life*, 18(1), 365–391. <https://doi.org/10.1007/s11482-022-10087-3>
- Xiao, J. J., Huang, J., Goyal, K., & Kumar, S. (2022). Financial capability: A systematic conceptual review, extension and synthesis. *International Journal of Bank Marketing*, 40, 1680–1717. <https://doi.org/10.1108/IJBM-05-2022-0185>
- Xiao, J. J., & Tao, C. (2020). Consumer finance/household finance: The definition and scope. *China Finance Review International*, 11, 1–25. <https://doi.org/10.1108/CFRI-04-2020-0032>
- Xu, M., & Eckstein, Y. (1995). Use of weighted least-squares method in evaluation of the relationship between dispersivity and field scale. *Groundwater*, 33, 905–908. <https://doi.org/10.1111/j.1745-6584.1995.tb00035.x>
- Zeqiraj, V., Sohag, K., & Hammoudeh, S. (2022). Financial inclusion in developing countries: Do quality institutions matter? *Journal of International Financial Markets Institutions and Money*, 81, Article 101677. <https://doi.org/10.1016/j.intfin.2022.101677>
- Zinman, J. (2015). Household debt: Facts, puzzles, theories, and policies. *Annual Review of Economics*, 7, 251–276. <https://doi.org/10.1146/annurev-economics-080614-115640>